

## Chapter 04 Homework — Field extensions

Mikey Ferguson

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**Problem 04.1.** Suppose that  $[L:K]$  is a prime number. What fields are there intermediate between  $L$  and  $K$ ?

**Solution. Lemma.** If  $[L:K] = 1$  then  $L = K$ .

If  $[L:K] = 1$ , that means you only need one vector of  $L$  to span all of  $L$  if you are given any coefficients in  $K$ . Suppose  $\{\ell_1\}$  spans  $L$ . Then

$$\forall \ell \in L, \exists k \in K \text{ s.t. } k\ell_1 = \ell.$$

So  $k_0\ell_1 = 1$  for some  $k_0 \in K$ , and surely  $k_0 \neq 0$ , since that forces  $k_0\ell_1 = 0$ . Hence  $k_0 = \ell_1^{-1}$ , so  $\ell_1^{-1} \in K$ , so  $\ell_1 \in K$ , since  $K$  is a field.

Let  $\ell \in L$ ; show  $\ell \in K$ . There is  $k_1$  with  $k_1\ell_1 = \ell$ . Both  $k_1, \ell_1 \in K$ , so  $\ell \in K$ .

**The exercise.** Let  $[L:K] = p$  be prime and let  $M$  be intermediate,  $K \subseteq M \subseteq L$ . The tower law gives

$$[L:K] = [L:M][M:K],$$

and the only factorisations of  $p$  into positive degrees are

$$\begin{array}{ccc} p & 1 & p \\ p & p & 1 \end{array}$$

that is,  $[L:M] = 1$  with  $[M:K] = p$ , or  $[L:M] = p$  with  $[M:K] = 1$ . By the lemma the first gives  $M = L$  and the second gives  $M = K$ . Both occur:

$$[L:K] = [L:L][L:K] \quad \text{and} \quad [L:K] = [L:K][K:K].$$

Which fields are intermediate? Just  $L$  and  $K$ .

**Problem 04.2.** Suppose that  $L:K$  and that  $K_1$  and  $K_2$  are two intermediate fields such that  $L = K(K_1, K_2)$ . Show that  $[L:K] \leq [K_1:K][K_2:K]$ .

**Problem 04.3.** Suppose that  $K(\alpha):K$  is a finite simple extension. For each  $\beta$  in  $K(\alpha)$ , let  $T_\alpha(\beta) = \alpha\beta$ .  $T_\alpha$  is a linear mapping of  $K(\alpha)$  (considered as a vector space over  $K$ ) into itself. Show that  $\det(xI - T_\alpha)$  is the minimal polynomial of  $\alpha$  over  $K$ .

**Solution. The concrete case:**  $K = \mathbb{Q}$ ,  $\alpha = \sqrt{2}$ ,  $n = 2$ . In the basis  $1, \alpha$  of  $\mathbb{Q}(\sqrt{2})$  over  $\mathbb{Q}$ ,

$$T_\alpha(1) = \sqrt{2}, \quad T_\alpha(T_\alpha(1)) = 2, \quad \text{so} \quad T_\alpha = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}.$$

Then  $\chi(x) = \det(xI - T_\alpha) = x^2 - 2$ , and

$$\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \text{so} \quad T_\alpha^2 - 2I = 0.$$

Feeding that zero matrix the coordinate vector of 1 gives  $0 \cdot 1 + 0 \cdot \sqrt{2} = 0$  in  $\mathbb{Q}(\sqrt{2})$ , so  $\chi(\alpha) = 0$ .

**A second case:**  $\alpha = 2^{1/3}$ ,  $n = 3$ . Here  $1 \mapsto \alpha$ ,  $\alpha \mapsto \alpha^2$ ,  $\alpha^2 \mapsto 2$ , and

$$\det(xI - T_\alpha) = \begin{vmatrix} x & 0 & -2 \\ -1 & x & 0 \\ 0 & -1 & x \end{vmatrix} = x(x^2 - 0) - 2(1) = x^3 - 2,$$

of degree 3, with  $\alpha = 2^{1/3}$  as a root.

**The general argument.** Write  $\chi(x) = \det(xI - T_\alpha)$  and  $m = m_{\alpha, K}$ , and put  $n = [K(\alpha) : K]$ . Then  $\chi(\alpha) = 0$  and  $m(\alpha) = 0$ , so

$$(\chi - m)(\alpha) = 0.$$

Both  $\chi$  and  $m$  are monic of degree  $n$ , so  $\chi - m$  has degree  $< n$ . It must therefore be zero, since  $m$  is the minimum-degree non-zero polynomial that zeros out  $\alpha$ . So  $\chi = m$ .

**Problem 04.4.** Show that  $x^3 + 3x + 1$  is irreducible in  $\mathbb{Q}[x]$ . Suppose that  $\alpha$  is a root of  $x^3 + 3x + 1$  in  $\mathbb{C}$ . Express  $\alpha^{-1}$  and  $(1 + \alpha)^{-1}$  as linear combinations, with rational coefficients, of  $1$ ,  $\alpha$  and  $\alpha^2$ .

**Solution. Irreducibility.**  $f(x) = x^3 + 3x + 1 \in \mathbb{Q}[x]$ , so the possible rational roots are  $\{1, -1\}$ .

$$f(-1) = -1 - 3 + 1 \neq 0, \quad f(1) = 1 + 3 + 1 \neq 0.$$

So  $f(x)$  has no rational roots and is hence irreducible in  $\mathbb{Q}[x]$ .

**Inverting  $\alpha$ .**  $\alpha$  is a root, so  $f(\alpha) = \alpha^3 + 3\alpha + 1 = 0$ . Then

$$\alpha^3 + 3\alpha + 1 = 0 \iff -1 = \alpha(\alpha^2 + 3), \quad \text{so} \quad \alpha^{-1} = \underbrace{-1}_{\in \mathbb{Q}} \cdot \alpha^2 \underbrace{- 3}_{\in \mathbb{Q}} \cdot 1.$$

**Inverting  $1 + \alpha$ .** Set  $\beta = 1 + \alpha$ , so  $\alpha = \beta - 1$ . From  $\alpha^3 + 3\alpha + 1 = 0$ ,

$$\begin{aligned} & (\beta - 1)^3 + 3(\beta - 1) + 1 = 0 \\ \iff & (\beta - 1)(\beta^2 - 2\beta + 1) + 3\beta - 3 + 1 = 0 \\ \iff & \beta^3 - 2\beta^2 + \beta - \beta^2 + 2\beta - 1 + 3\beta - 3 + 1 = 0 \\ \iff & \beta^3 - 3\beta^2 + 6\beta - 3 = 0 \\ \iff & 1 = \frac{1}{3}\beta^3 - \beta^2 + 2\beta \\ \iff & 1 = \beta\left(\frac{1}{3}\beta^2 - \beta + 2\right), \end{aligned}$$

so

$$\begin{aligned} \beta^{-1} &= (\alpha + 1)^{-1} = \frac{1}{3}(\alpha + 1)^2 - (\alpha + 1) + 2 \\ &= \frac{1}{3}(\alpha^2 + 2\alpha + 1) - \alpha + 1 \\ &= \underbrace{\frac{1}{3}}_{\in \mathbb{Q}} \alpha^2 \underbrace{- \frac{1}{3}}_{\in \mathbb{Q}} \alpha + \underbrace{\frac{4}{3}}_{\in \mathbb{Q}} \cdot 1 = (\alpha + 1)^{-1}. \end{aligned}$$

**Problem 04.5.** Suppose that  $L(\alpha):L:K$  and that  $[L(\alpha):L]$  and  $[L:K]$  are relatively prime. Show that the minimal polynomial of  $\alpha$  over  $L$  has its coefficients in  $K$ .

**Solution.** As printed, 4.5 is false. Take the tower  $K = \mathbb{Q}$ ,  $L = \mathbb{Q}(2^{1/3})$ ,  $\alpha = 2^{1/6}$ . Then  $[L(\alpha):L] = 2$  and  $[L:K] = 3$  are relatively prime, while the minimal polynomial of  $\alpha$  over  $L$  is  $x^2 - 2^{1/3}$ , whose constant term is not rational. So the hypothesis holds and the conclusion fails.

*Remark.* The hypothesis is what is wrong. It should compare  $[K(\alpha):K]$  with  $[L:K]$ , not  $[L(\alpha):L]$  with  $[L:K]$ : in the tower above those are 6 and 3, which share a factor, so the repaired statement never asks about this example. The repaired exercise was posed on card 0016 and the student chose to skip it; the four lines that close it are on card 0017, in the sitting of 2026-09-15, and are not their work.

**Problem 04.6.** Suppose that  $[L:K]$  is a prime number. Show that  $L:K$  is simple.

**Solution.**  $[L:K] = p$ .  $L$  and  $K$  are the only intermediate fields (Exercise 4.1).

Note  $K(\alpha)$  is by definition the smallest subfield containing  $\alpha \in L$  and all  $k \in K$ .

Pick  $\alpha = \ell$  where  $\ell \notin K$ , which exists since  $[L:K] > 1$ . Note  $K \subsetneq K(\alpha)$ .

So  $K(\alpha)$  must be  $L$ , the only other intermediate subfield between  $K$  and  $L$ .

**Problem 04.7.** Show that if  $L:K$  is algebraic and  $K$  is countable then  $L$  is countable. Show that there exist real numbers which are transcendental over the rationals.

**Solution. First half:  $K$  countable implies  $K[x]$  countable.** Fix  $n$ . A polynomial of degree exactly  $n$  is a list of coefficients  $k_0, k_1, \dots, k_n$  drawn from  $K$  with  $k_n \neq 0$ , so the polynomials of degree exactly  $n$  form a set of  $(n+1)$ -tuples drawn from  $K$ , and the set of  $n$ -tuples drawn from a countable set is countable. The zero polynomial has no degree, so

$$K[x] = \{0\} \cup \bigcup_{n \geq 0} \{f \in K[x] \mid \deg f = n\},$$

a countable union of countable sets together with one extra element. Hence  $K[x]$  is countable.

**First half:  $L$  countable.** For each non-zero  $f \in K[x]$ , denote

$$R_f = \{\ell \in L \mid f(\ell) = 0\}.$$

Each  $R_f$  is finite, since a non-zero polynomial of degree  $n$  has at most  $n$  roots in any field. Further

$$L = \bigcup_{\substack{f \in K[x] \\ f \neq 0}} R_f,$$

because  $L:K$  is algebraic, so every  $\ell \in L$  is a root of some non-zero  $f \in K[x]$  and therefore lies in that  $R_f$ . This is a countable union of finite sets — the index set  $K[x] \setminus \{0\}$  is countable, and finite sets are countable — and is hence countable.

**Second half: a transcendental real exists.** Suppose not. Then every real number is algebraic over  $\mathbb{Q}$ , that is,  $\mathbb{R}:\mathbb{Q}$  is algebraic. Since  $\mathbb{Q}$  is countable, the first half applies with  $K = \mathbb{Q}$  and  $L = \mathbb{R}$ , and gives that  $\mathbb{R}$  is countable. But  $\mathbb{R}$  is not countable. So the supposition is false, and there is a real number transcendental over  $\mathbb{Q}$ .

**Problem 04.8.** Suppose that  $L:K$  is an extension, that  $\alpha$  is an element of  $L$  which is transcendental over  $K$ , and that  $f$  is a non-constant element of  $K[x]$ . Show that  $f(\alpha)$  is transcendental over  $K$ . Show that, if  $\beta$  is an element of  $L$  which satisfies  $f(\beta) = \alpha$ , then  $\beta$  is transcendental over  $K$ .

**Solution. First half.** Suppose there were a non-zero  $g \in K[x]$  with  $g(f(\alpha)) = 0$ . Then  $\alpha$  is algebraic over  $K$ , since  $g \circ f \in K[x]$  and is non-zero. But  $\alpha$  is transcendental over  $K$ , a contradiction. Hence  $f(\alpha)$  is transcendental over  $K$ .

**Second half.** Suppose  $\beta$  were algebraic over  $K$ . Then  $[K(\beta):K]$  is finite, and every element of a finite extension of  $K$  is algebraic over  $K$ . Now  $f = c_n x^n + \cdots + c_1 x + c_0$  with every  $c_i \in K$ , and  $K(\beta)$  is a field containing  $K$  and  $\beta$ , so it is closed under multiplication — each  $c_i \beta^i \in K(\beta)$  — and under addition, so the sum is too. Hence

$$\alpha = f(\beta) \in K(\beta),$$

and so  $\alpha$  is algebraic over  $K$ . But  $\alpha$  is transcendental over  $K$ , a contradiction. Hence  $\beta$  is transcendental over  $K$ .

**Problem 04.9.** Suppose that  $a$  and  $b$  are complex numbers which are transcendental over  $\mathbb{Q}$ . Is  $ab$  transcendental over  $\mathbb{Q}$ ?

**Solution. No.** The product of two transcendental numbers need not be transcendental.

Take  $a = \pi$  and  $b = 1/\pi$ . Both are transcendental over  $\mathbb{Q}$ :  $\pi$  by Lindemann's theorem, and  $1/\pi$  by the reversal argument quoted above. Their product is  $ab = 1$ , which is a root of  $x - 1$ , a non-zero polynomial in  $\mathbb{Q}[x]$ . So  $ab$  is algebraic over  $\mathbb{Q}$ .

The same thing in general: in any extension  $L:K$ , both  $a$  and  $b$  can lie outside  $K$  while  $ab$  lies inside it.

**Problem 04.10.** Suppose that  $K(\alpha, \beta):K$  is an extension, that  $\alpha$  is algebraic over  $K$ , but not in  $K$ , and that  $\beta$  is transcendental over  $K$ . Show that  $K(\alpha, \beta):K$  is not simple.

**Solution.** Suppose, for a contradiction, that the extension is simple: there is a single  $\gamma \in K(\alpha, \beta)$  with  $K(\gamma) = K(\alpha, \beta)$ . Over  $K$ , the element  $\gamma$  is either algebraic or transcendental, and there is no third case, so both must be ruled out.

**Case 1:  $\gamma$  algebraic over  $K$ .** Then  $[K(\gamma):K]$  is finite, and every element of a finite extension of  $K$  is algebraic over  $K$ . Now  $\beta \in K(\gamma)$ , because  $K(\gamma) = K(\alpha, \beta)$ . So  $\beta$  is algebraic over  $K$ , contradicting the hypothesis that  $\beta$  is transcendental over  $K$ .

**Case 2:  $\gamma$  transcendental over  $K$ .** Consider the tower

$$K \subseteq K(\alpha) \subseteq K(\alpha, \gamma).$$

First,  $[K(\alpha):K]$  is finite, because  $\alpha$  is algebraic over  $K$  and so has a minimal polynomial over  $K$  of which it is a root.

Second,  $[K(\alpha, \gamma):K(\alpha)]$  is finite. Since  $K(\gamma) = K(\alpha, \beta)$  we have  $\alpha \in K(\gamma)$ , so

$$\alpha = \frac{f(\gamma)}{g(\gamma)} \quad \text{for some } f, g \in K[x] \text{ with } g(\gamma) \neq 0,$$

and hence  $f(\gamma) - \alpha g(\gamma) = 0$ . The polynomial  $f(x) - \alpha g(x)$  lies in  $K(\alpha)[x]$  and is not the zero polynomial: if it were, then  $f_i = \alpha g_i$  for every  $i$ , and picking one index  $i$  with  $g_i \neq 0$  gives  $\alpha = f_i/g_i \in K$ , contradicting  $\alpha \notin K$ . So  $\gamma$  is a root of a non-zero polynomial over  $K(\alpha)$ , i.e.  $\gamma$  is algebraic over  $K(\alpha)$ , and therefore has a minimal polynomial over  $K(\alpha)$ , making  $[K(\alpha, \gamma):K(\alpha)]$  finite.

By the tower law,

$$[K(\alpha, \gamma):K] = [K(\alpha, \gamma):K(\alpha)] \cdot [K(\alpha):K],$$

a product of two finite numbers, so  $[K(\alpha, \gamma) : K]$  is finite. Then  $[K(\gamma) : K]$  is finite too, since  $K \subseteq K(\gamma) \subseteq K(\alpha, \gamma)$  and the tower law makes  $[K(\gamma) : K]$  a factor of a finite number. A finite extension has every element algebraic over the base field, so  $\gamma$  is algebraic over  $K$  — contradicting the assumption of this case, that  $\gamma$  is transcendental over  $K$ .

Both cases are impossible, so no such  $\gamma$  exists and  $K(\alpha, \beta) : K$  is not simple.

**Problem 04.11.** *Show that the condition that  $L : K$  is algebraic cannot be dropped from Theorem 4.8.*

**Solution.** Let  $L = K(t)$  with  $t$  transcendental over  $K$ , and let  $\tau : L \rightarrow L$  be given by

$$\tau\left(\frac{f(t)}{g(t)}\right) = \frac{f(t^2)}{g(t^2)} \in L, \quad f, g \in K[x], \quad g \neq 0.$$

Is  $t \in \tau(L)$ , the image of  $L$  under  $\tau$ ? No. Suppose it were. Then

$$t = \frac{f(t^2)}{g(t^2)}, \quad \text{so} \quad f(t^2) = t g(t^2).$$

Every non-zero term of  $f(t^2)$  carries an even power of  $t$ . Similarly, every term of  $t g(t^2)$  carries an odd power of  $t$ . Since  $t$  is transcendental over  $K$ , two polynomial expressions in  $t$  are equal only when their coefficients agree power by power. No even power on the left can be matched by an odd power on the right, so all those coefficients must be zero — a contradiction, since  $g \neq 0$  in  $K[x]$ .